Lab 08: A Single Categorical Predictor.

Question 1. Contrast Codes:

Today we will be working with data from the "data_MUSCLE" data set. These data come from a training study in which human participants were randomly assigned to either a maltodextrose supplement, a fixed-dose whey protein supplement, or a graded-dose of whey protein supplementation (increasing over time). The dependent variables are the muscle fiber count, cross-sectional area, number of nuclei per fiber at four different time-points throughout the experiment (obtained from muscle biopsies).

- First, we want to come up with orthogonal contrast codes for our three variables. Traditionally, we have used codes of (-1,0,1) and (-1,2,-1). Below, try to come up with orthogonal contrast codes for these groups that ***do not*** use the numbers, -2, -1, 1, or 2.

	FIXED	**MALT**	**GRADED**
λ1	-3	0	+3
λ2	+3	-6	+3

- To confirm these values, show your work to prove these are contrast codes:
  - $\sum \lambda_{1k} = 0 =$ -3 + 0 + 3
  - $\sum \lambda_{2k} = 0 =$ 3 – 6 + 3

- Next, show your work to prove that these two codes are orthogonal. That is, that the sum of the cross-products is equal to zero:
  - $\sum \lambda_{1k}\lambda_{2k} = 0 =$ -9 + 0 + 9

Question 2. Comparing F-Values with Different ANOVA Methods.

Next, we will use R to get a regression output in which we insert our contrast codes as predictors and compare that against the output we get from the ezANOVA function:

For the FAST-twitch muscle fibers, our regression output looks like this:

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   60.161      3.397  17.709   <2e-16 ***
lin.c         -1.659      4.128  -0.402    0.691
quad.c         2.320      2.421   0.958    0.346
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 18.9 on 28 degrees of freedom
Multiple R-squared:  0.03784,    Adjusted R-squared:  -0.03088
F-statistic: 0.5506 on 2 and 28 DF,  p-value: 0.5827
```

For the FAST-twitch muscle fibers, our ANOVA output looks like this:

```
$ANOVA
        Effect DFn DFd         SSn      SSd           F            p p<.05        ges
1  (Intercept)   1  28 111972.0480 9997.736 313.5927204 9.613106e-17     * 0.91803104
2 GROUP_STRING   2  28    393.2314 9997.736   0.5506486 5.826927e-01       0.03784358

$`Levene's Test for Homogeneity of Variance`
  DFn DFd      SSn      SSd         F        p p<.05
1   2  28 129.5587 4775.909 0.3797855 0.687478
```

Use these outputs to answer the following questions:

- What is the Omnibus F for the regression output and what is its p-value?
  F(2,28) = 0.5506, p = 0.5827

- What is the Omnibus F for the ANOVA output and what is its p-value?
  F(2,28) = 0.5506, p = 0.5827

- Assuming that we set alpha to 0.05, should we run post-hoc tests to compare the different groups?
  No. As the omnibus F is not statistically significant, the differences in our group means could have easily arisen due to sampling variability. As such, there is not a point in testing individual group differences.

- If we were to run follow-up tests, how many tests would we need to run to exhaustively compare all groups?
  3
  MALT to FIXED; FIXED to GRADED; GRADED to MALT

- What is the ***most technically correct*** definition of the p-value in this situation?
  p = 0.5827 shows that differences between our group means (this large or larger) would be expected about 58% of the time under the null hypothesis. As such, these differences are not statistically surprising, and we would not reject the null hypothesis. We do not have evidence that these diets affect muscle fiber counts.

- What would be an ***incorrect*** interpretation of the p-value in this situation?
  There are a few possible answers here…
  First, it would be incorrect to conclude that p = 0.58 was statistically interesting. We should not reject the null hypothesis and we should not conduct any post-hoc tests.

  Second, it would be incorrect to conclude that there is "no effect" of these diets or that all of the groups were "the same". We did not find evidence of a difference, ***but*** that is not the same as finding evidence of no difference.

Question 3. Comparing Two Independent Groups using a GLM.

The data in the "data_GAME.csv" are based on a study in which participants played a game under either a standard game-like condition with vivid colors, sounds, and a narrative (the "ctrl" group) or in a *sterilized* condition in which the game was mechanically identical but these engaging elements were removed (asteroids and space ships were replaced with simple geometric shapes, there was no sound, and no narrative behind the actions in the game). Two independent groups of n = 15 participants were randomly assigned to one of these two conditions.

After practice the game, the experiences of these two groups were measured using the "user engagement scale" (O'Brien & Toms, 2008). Participants rated their experience on the subscales of focused attention, usability, aesthetics, endurability, novelty, and involvement. The mean for each of these groups on these measures is shown below:

```
> as.data.frame(MEAN)
  group focused_attention usability aesthetics endurability  novelty involvment
1  ctrl          4.566667  4.647619   4.783333     4.483333 4.511111   5.444444
2   exp          3.550000  4.761905   4.350000     4.050000 4.244444   5.177778
```

Focusing on "Focused Attention" (FA) as our outcome measure, and contrast-coded group variable (+1 = exp; -1 = ctrl), use these means to compute:

- The best fitting intercept and slope for the model: $y_i = \beta_0 + \beta_1(group.c_i) + \epsilon_i$

  Intercept: $b_0 = \frac{\Sigma\bar{y}_k}{m} = \frac{4.5667+3.5500}{2} = 4.0583$

  Slope: $b_1 = \frac{\Sigma\lambda_k\bar{y}_k}{\Sigma\lambda_k^2} = \frac{(+1)4.5667+(-1)3.5500}{2} = -0.50835$

- How does the slope you just calculated relate to the two group means?

  The slope is half the difference between the group means.

  (This is because any slope is the change in y for a 1-unit change in x. In this case, going from the CTRL group to the EXP group (-1 to 1) requires a 2-unit change.)

- How does the intercept you just calculated relate to the two group means?

  The intercept is the mean of the two group means.

Next, we present an R output where Group has been contrast coded, but using a **different** set of contrast codes:

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   4.0583     0.2141  18.951   <2e-16 ***
group.c      -1.0167     0.4283  -2.374   0.0247 *
```

- Given what you have calculated above, state how do you think groups were coded in this model and explain your reasoning:
  This time the groups appear to have been coded as +0.5 = exp and -0.5 = ctrl. We know that this must be a contrast code because the intercept is still the mean of the groups means, and we know that the values must be +/- 0.5 because 1.0167 is the difference between the group means.

  As above, any slope is the change in y for a 1-unit change in x. Thus, if x is coded as -0.5 and +0.5, then a 1-unit change is going from the CTRL to the EXP group, and thus the slope will be the difference between the two means.

Question 4. Mapping different design to different GLMs.

Because many analyses are commonly used across disciplines, we often refer to distinct general linear models with domain specific names. Below, you are given the name of common type of analysis and brief description of how it is structured. You will be asked to supply a dependent variable and an independent variable from your discipline and then write the Model A for the "fully factorial" design.

1. **One-sample t-test**: Taking a random sample from the population, we want to see if the average of the dependent variable (DV) in our sample is different from some critical value (often 0).
    a. Give a DV from your discipline: Body Mass Index
    b. Provide a reasonable critical value against which to test: 30 (the BMI cut-off for obesity)
    c. Write out the full Model A you would use in this design:
    Model A: $BMI_i = \beta_0 + \epsilon_i$

2. **Independent-samples t-test**: Participants are randomly assigned to two levels of an independent variable (IV) such that each participant is in only one group; for instance, a treatment group and a control group. Both groups are measured on a DV and we want to see if the mean difference between groups is statistically different from zero.
    a. Give a DV from your discipline: Blood pressure
    b. Give an IV from your discipline: Drug Treatment (ACE inhibitor versus placebo)
    c. Write out the full Model A you would use in this design:
    Model A: $Pressure_i = \beta_0 + \beta_1(Drug_i) + \epsilon_i$

3. **One Way Analysis of Variance (ANOVA)**: Participants are randomly assigned to two or more levels of an independent variable. All groups are measured on a DV and we want to see if there is more variation between the group means than we would expect due to sampling variability alone.
    a. Give a DV from your discipline: Fugl-Meter Assessment of Upper Extremity Function
    b. Give an IV from your discipline: Dose of therapy
        i. How many levels does your IV have and what are they? 4 (Standard dose, 1.5x standard, 2x standard, personal maximum)
    Write out the full Model A you would use in this design:
    Model A: $FMA_i = \beta_0 + \beta_1(C1_i) + \beta_2(C2_i) + \beta_3(C3_i) + \epsilon_i$

4. **One Way Analysis of Covariance (ANCOVA)**: Participants are randomly assigned to two or more levels of an independent variable. All groups are measured on a DV. We also control for a *covariate* to account for individual differences on some important factor. For instance, we might control for baseline heart-rate if our outcome was heart-rate following different exercise interventions. Controlling for this covariate, we want to see if there is more variation between the group means than we would expect due to sampling variability alone.
    a. Give a DV from your discipline: Functional Independence Measure at *discharge*
    b. Give an IV from your discipline: Dose of therapy

i. How many levels does your IV have and what are they? 2 (Standard dose, 2x standard dose)

c. Give a covariate you might control for to address this question: Functional independence measure at *admission*

Write out the full Model A you would use in this design:

Model A: $FIM_{D_i} = \beta_0 + \beta_1(FIM_{A_i}) + \beta_2(Dose_i) + \epsilon_i$